

Convert the expression

$A * B - C / D + E$ to prefix notation.

$$= (A * B) - (C / D) + E$$

The traversal should be from left to right always

$$\begin{aligned}
 &= (\star AB) - (C|D) + E \\
 &= (\star AB) - (|CD) + E \\
 &= (T_1 - T_2) + E \\
 &= -T_1 T_2 + E \\
 &= (T_3 + E) \\
 &= +T_3 E \\
 &= + -T_1 T_2 E \\
 &= + - \star AB T_2 E \\
 &= \boxed{+ - \star AB | CD E}
 \end{aligned}
 \left| \begin{array}{l} T_1 \\ = \star AB \\ T_2 \\ = |CD \\ = -T_1 T_2 \\ \quad +/ \\ \quad T_3 \end{array} \right.$$

② Convert the expression

$(A+B) / C * D - E$ into
prefix notation

$$(A+B) / C * D - E$$

$$(+AB) / C * D - E$$

$$(T_1 / C) * D - E$$

$$(/T_1 C) * D - E$$

$$(T_2 * D) - E$$

$$(*T_2 D) - E$$

$$T_3 - E$$

$$-T_3 E$$

$$- * T_2 D E$$

$$- * / T_1 C D E$$

$$\boxed{- * / + A B C D E}$$

$$+AB = T_1$$

$$/T_1 C = T_2$$

$$*T_2 D = T_3$$

Conversion of infix to postfix

Rules:

① Priorities of operators:

$\wedge$ = highest priority

$*, /$ = next priority

$+, -$ = lowest priority

② No two operators of same priority can stay together in stack column.

③ Lowest priority operator cannot be placed before highest priority.

④ Any operator placed with $($, should be popped out.

place operand $\rightarrow$ postfix

operator $\rightarrow$ stack

$(A+B)C*(D+E)-F$

<u>Symbol</u>	<u>Stack</u>	<u>postfix</u>
$($	$($	
A	$($	A
$+$	$(+$	A
B	$(+$	AB
$/$	$(+ /$	AB
C	$(+ /$	ABC

*

C + / *

ABC

*

C + *

ABC /

(

(+ * C

ABC /

)

(+ * C

ABC / D

+

(+ * (+

ABC / D

E

(+ * (+

ABC) DE

)

(+ * (+)
pop

ABC) DE

-

(+ * -
pop

ABC) DE +

-

(+ -
pop

ABC / DE + *

F

(-

ABC / DE + * +

)

(-)
pop

ABC / DE + * + F

ABC / DE + * + F -

↓
output

Evaluate the postfix notation

5, 6, 2, +, *, 12, 4, /, -

<u>Symbol</u> <u>Scanned</u>	<u>Stack</u>
5	5
6	5, 6
2	5, 6, 2
+	5, 8
*	40
12	40, 12
4	40, 12, 4
/	40, 3
-	37 Result

Evaluate the prefix expression

+, -, *, 2, 2, /, 16, 8, 5

Reverse the prefix notation

5, 8, 16, /, 2, 2, *, -, +

Symbol	Scanned	Stack
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5

5

8

5, 8

16

5, 8, 16

/

5, 2

2

5, 2, 2

2

5, 2, 2, 2

*

5, 2, 4

-

5, 2

+

7 result

FE